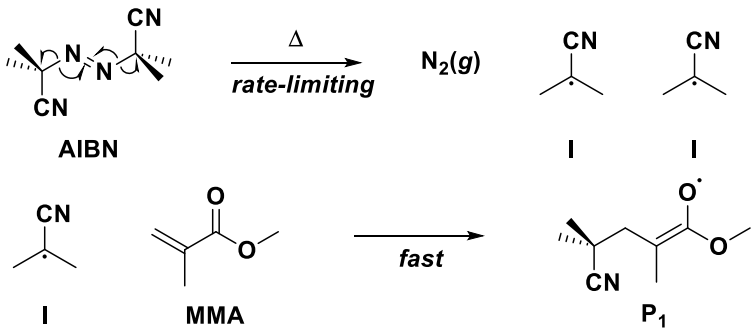
ID number: _____
Cumulative Exam 10-15-2024
Organic Division

1. Shown below are the elementary steps for free radical polymerization of methyl methacrylate (MMA). Study this mechanism. Then, using single-barbed arrows, draw arrows to show the transfer of electrons in each step. Then, write the rate law for each step where prompted.

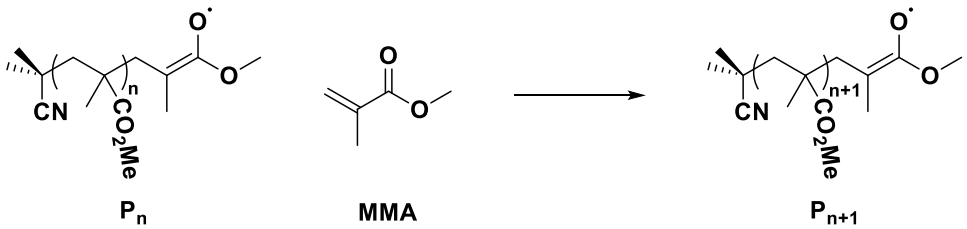
Free Radical Polymerization

Initiation



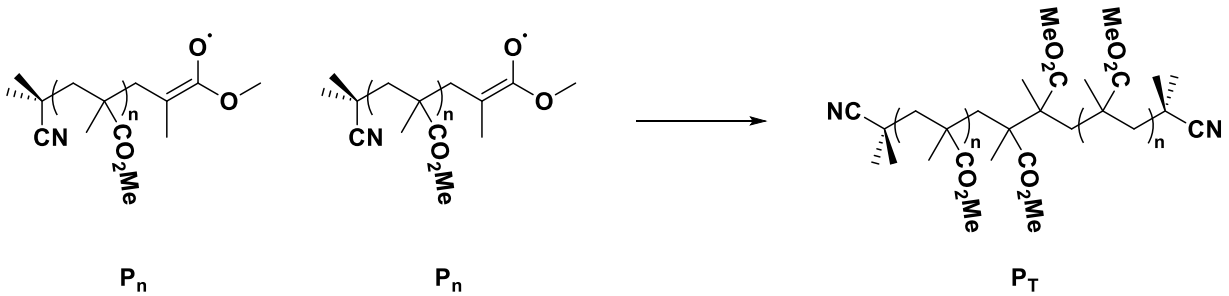
Rate = k_i _____

Propagation



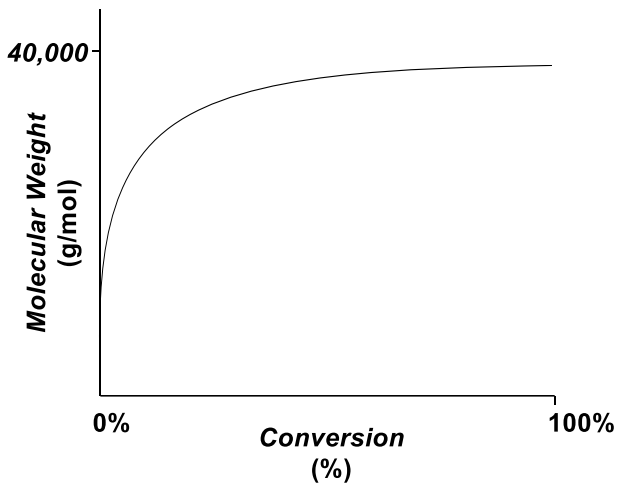
Rate = k_p _____

Termination



Rate = k_T _____

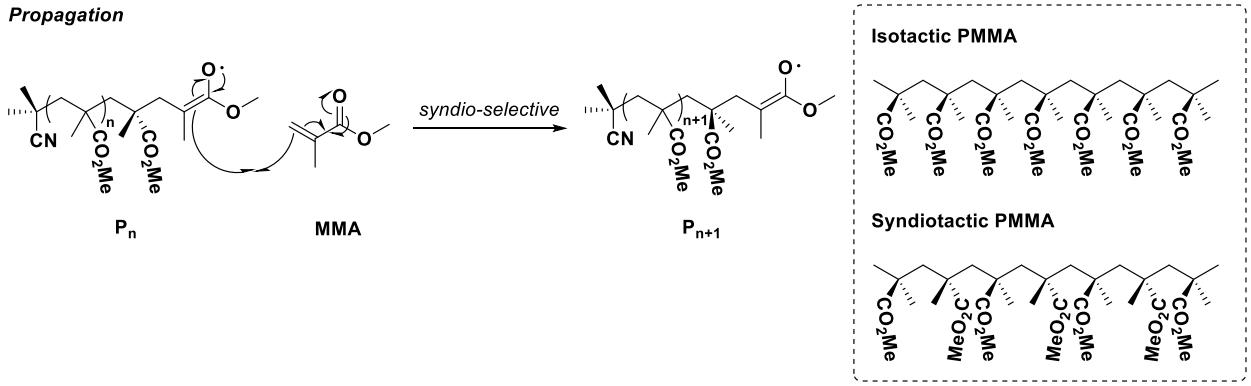
2. Below is the Molecular Weight vs Conversion plot for polymers generated through the free radical polymerization mechanism shown in question 1. Molecular weight is the average size of each chain in (g/mol). Conversion is measured in terms of monomer depletion (*e.g.* 50 % conversion means half the monomer has been depleted, 100 % conversion means the monomer is completely gone). Rationalize the shape of this curve by using arguments based on the mechanism.



Answer:

3. Tacticity is a way to describe the stereochemistry of polymers by comparing the relationship of each monomer unit to its neighboring unit. Isotactic poly(methyl methacrylate) (PMMA) is defined as a chain where all the stereogenic carbons along the backbone are uniform. Syndiotactic PMMA is defined by stereogenic carbons along the backbone being opposite that of their neighboring units. While PMMA produced through free radical polymerization is generally described as atactic (or being stereo-random), there is actually a slight bias towards the syndiotactic structure, with most chains being ~70 % syndiotactic. Using the propagation mechanism provided below, identify the source of this stereo-selectivity. In other words, where does this slight preference for syntiotacticity come from?

Hint: To generate stereo-selectivity, there must be an asymmetric transition state.



Answer:

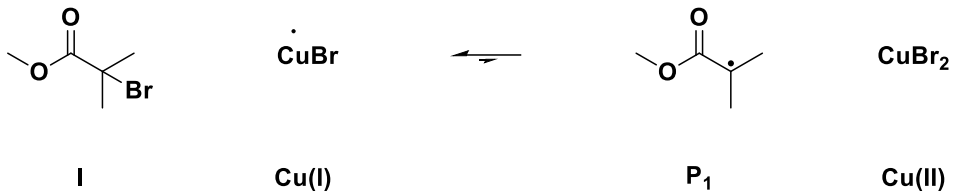
4. In recent years, more sophisticated ways of polymerizing methacrylates have been developed. One such technique shown on the next page is Atom Transfer Radical Polymerization (ATRP). This method contains an additional step (activation/deactivation) wherein Cu(II) can reversibly quench the unstable propagating radical (transient radical) to form a dormant alkyl halide species and more stable Cu(I) radical (persistent radical), thus greatly reducing the number of propagating radicals at any given time. This is known as the persistent radical effect. Study this mechanism. Then, using single-barbed arrows, draw arrows to show the transfer of electrons in each step. Then, write the rate law for each step where prompted.

5. While ATRP propagation is generally slower than free radical polymerization in terms of propagation rate, one advantage of ATRP is that termination is either extremely rare, or totally unobservable. Using the rate laws of each elementary step, explain how the persistent radical effect prevents termination while still allowing propagation to proceed.

Answer:

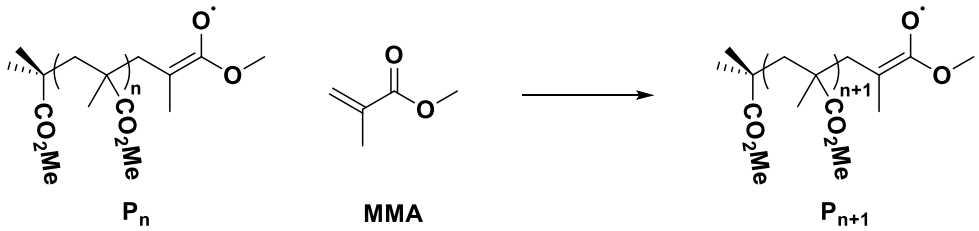
Atom Transfer Radical Polymerization (ATRP)

Initiation



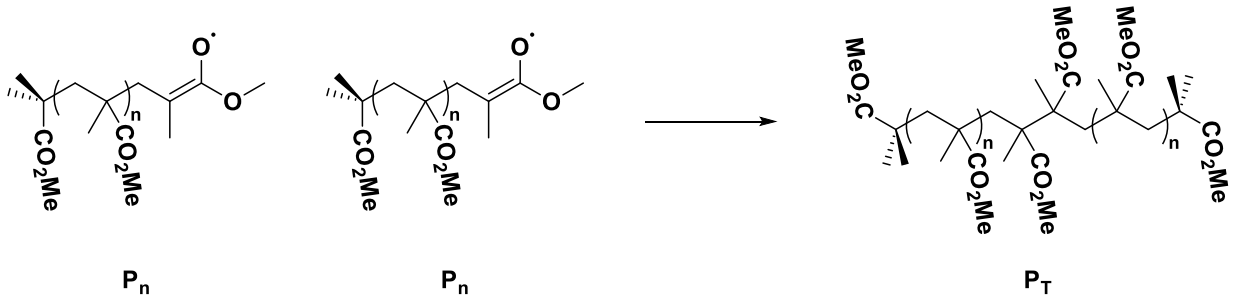
Rate = k_i _____

Propagation



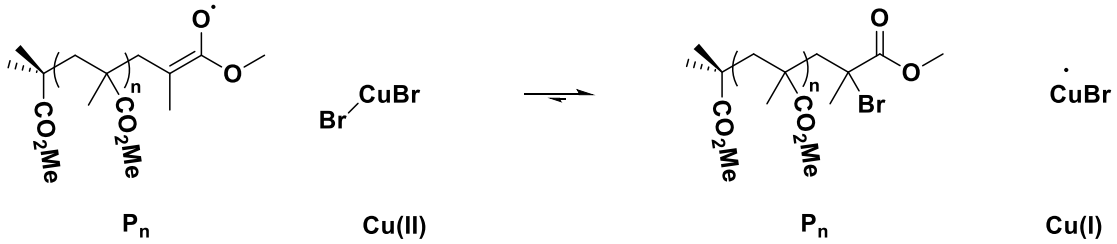
Rate = k_p _____

Termination

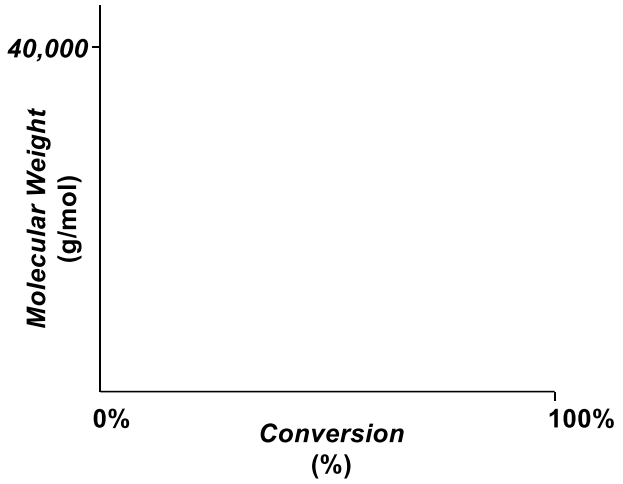


Rate = k_T _____

Activation/Deactivation



6. Using the plot below, draw in the expected shape of the molecular weight vs conversion curve based on the mechanism of ATRP and the assumption that termination is not observed.



7. One of the advantages of ATRP is that the average polymer molecular weight can be predicted and manipulated by experimental conditions. What experimental conditions could you change in order to target a specific polymer molecular weight?

   Answer: